

Chapter 11

Parameter List

11.1 Parameter Group Description

Parameter access address: index+subindex, both in hexadecimal format

The CiA402 protocol has the following constraints on the address of system parameters.

Index	Description
0001h—0FFFh	Data type description
1000h—1FFFh	CoE communication object
2000h—5FFFh	Manufacturer specific object
6000h—9FFFh	Sub-protocol object
A000h—FFFFh	Reserved

11.2 Parameter List

11.2.1 Common Parameters in Group 2000h

Parameters (2000h/C00)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	C00.00	Control mode	10: EtherCAT	0-10	10	-	U16	At stop	Immediately
02h	C00.01	Motor rotating direction	0: CCW 1: CW	0-1	0	-	U16	At stop	Upon re-power-on
05h	C00.04	Auto-tuning mode	0: Manual mode 1: Standard mode 2: Positioning mode	0-2	1	-	U16	During operation	Immediately
06h	C00.05	Stiffness level	-	1-31	12	-	U16	During operation	Immediately
07h	C00.06	Load inertia ratio	-	0-12000	100	%	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
08h	C00.07	Absolute mode	0: Incremental position mode 1: Absolute position linear mode 2: Absolute position linear infinite mode 3: Absolute position single-turn mode 4: Absolute position rotation mode 5: Absolute mechanical single-turn mode (operating direction selectable)	0-5	0	-	U16	At stop	Upon re-power-on
11h	C00.10	Bleeder resistor selection	0: Internal bleeder resistor 1: External bleeder resistor 2: No bleeder resistor 3: Capacitor bleeder resistor	0-3	0	-	U16	At stop	Immediately
12h	C00.11	Bleeder resistor power	-	1-65535	50	W	U16	At stop	Immediately
13h	C00.12	Bleeder resistor resistance	-	1-65535	50	Ω	U16	At stop	Immediately
14h	C00.13	Bleeder resistor heat dissipation coefficient	-	1-100	30	-	U16	During operation	Immediately
15h	C00.14	Brake enable switch	-	0-1	0	-	U16	At stop	Immediately
17h	C00.16	Panel display	0: Default display 1: Speed display 2: Torque display 3: Voltage display 4: Load rate display	0-4	0	-	U16	During operation	Immediately
32h	C00.31	Super user	-	0-65535	0	-	U16	During operation	Immediately

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- For details about parameters above, refer to section 11.3.1 "Group C00".

Basic Gain Parameters (2001h/C01)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	C01.00	1st position loop gain	-	0-20000	400	0.1rad/s	U16	During operation	Immediately
02h	C01.01	1st speed loop gain	-	1-20000	250	0.1Hz	U16	During operation	Immediately
03h	C01.02	1st speed loop integral time parameter	-	1-51200	3184	0.01ms	U16	During operation	Immediately
04h	C01.03	1st torque reference filter cutoff frequency	-	5-16000	200	Hz	U16	During operation	Immediately
09h	C01.08	2nd position loop gain	-	0-20000	560	0.1rad/s	U16	During operation	Immediately
0Ah	C01.09	2nd speed loop gain	-	1-20000	350	0.1Hz	U16	During operation	Immediately
0Bh	C01.0A	2nd speed loop integral time parameter	-	1-51200	2274	0.01ms	U16	During operation	Immediately
0Ch	C01.0B	2nd torque reference filter cutoff frequency	-	5-16000	280	Hz	U16	During operation	Immediately
11h	C01.10	Speed feedback filter	0: Internal setting 1: Low-pass filter 2: Overlapping average filter 3: Speed observer 4: No filter	0-4	0	-	U16	At stop	Immediately
12h	C01.11	Cutoff frequency of speed feedback low-pass filter	-	10-16000	8000	Hz	U16	During operation	Immediately
13h	C01.12	Speed feedback overlapping average filter time constant	0: No filter 1: 2 times filter 2: 4 times filter 3: 8 times filter 4: 16 times filter 5: 32 times filter 6: 64 times filter	0-6	0	-	U16	During operation	Immediately
14h	C01.13	Speed feedforward source	0: No feedforward 1: Internal reference 2: Model tracking 5: Communication	0-5	0	-	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
15h	C01.14	Speed feedforward percentage	-	0-2000	0	0.1%	U16	During operation	Immediately
16h	C01.15	Speed feedforward filter cutoff frequency	-	5-16000	318	Hz	U16	During operation	Immediately
17h	C01.16	Torque feedforward source	0: No feedforward 1: Internal reference 2: Model tracking 5: Communication	0-5	0	-	U16	During operation	Immediately
18h	C01.17	Torque feedforward percentage	-	0-2000	0	0.1%	U16	During operation	Immediately
19h	C01.18	Torque feedforward filter cutoff frequency	-	5-16000	318	Hz	U16	During operation	Immediately
1Ch	C01.1B	PDF control coefficient	-	0-1000	1000	0.1%	U16	During operation	Immediately
1Dh	C01.1C	Damping factor control coefficient	-	0-1000	0	0.1%	U16	During operation	Immediately
21h	C01.20	Position reference overlapping average filter time constant A	-	0-1280	0	0.1ms	U16	At stop	Immediately
22h	C01.21	Position reference overlapping average filter time constant B	-	0-1280	0	0.1ms	U16	At stop	Immediately
23h	C01.22	Position reference low-pass filter time constant A	-	0-65535	0	0.1ms	U16	At stop	Immediately
24h	C01.23	Position reference low-pass filter time constant B	-	0-65535	0	0.1ms	U16	At stop	Immediately
25h	C01.24	1st notch filter frequency of position reference	-	0-2000	0	0.1Hz	U16	At stop	Immediately
26h	C01.25	1st notch filter width of position reference	-	0-1000	0	0.1%	U16	At stop	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
27h	C01.26	1st notch filter depth of position reference	-	10-1000	1000	0.1%	U16	At stop	Immediately
28h	C01.27	2nd notch filter frequency of position reference	-	0-2000	0	0.1Hz	U16	At stop	Immediately
29h	C01.28	2nd notch filter width of position reference	-	0-1000	0	0.1%	U16	At stop	Immediately
2Ah	C01.29	2nd notch filter depth of position reference	-	10-1000	1000	0.1%	U16	At stop	Immediately
2Bh	C01.2A	Position reference pre-charge filter time constant	-	0-1280	0	0.1ms	U16	At stop	Immediately
31h	C01.30	Adaptive notch mode	0: Disabled 1: 1st notch 2: 2nd notch 3: Notch parameter reset 4: Resonance frequency tested only	0-4	0	-	U16	During operation	Immediately
32h	C01.31	Adaptive notch test times	-	0-65535	0	Times	U16	At stop	Immediately
39h	C01.38	Gain switchover mode	0: Fixed to the 1st gain set 1: DI switchover 2: DI P-PI switchover 3: Torque reference 4: Speed reference 5: Speed feedback 6: Speed reference change rate 7: Position deviation 8: Position reference	0-8	0	-	U16	At stop	Immediately
3Ah	C01.39	Gain switchover time	-	10-10000	50	0.1ms	U16	During operation	Immediately
3Bh	C01.3A	Gain switchover threshold	-	0-65535	10	-	U16	During operation	Immediately
3Ch	C01.3B	Gain switchover loop width	-	0-65535	10	-	U16	During operation	Immediately
41h	C01.40	Frequency of the 1st notch	-	10-8000	8000	Hz	U16	During operation	Immediately
42h	C01.41	Width level of the 1st notch	-	0-4000	0	0.1%	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
43h	C01.42	Depth level of the 1st notch	-	10-1000	1000	0.1%	U16	During operation	Immediately
44h	C01.43	Frequency of the 2nd notch	-	10-8000	8000	Hz	U16	During operation	Immediately
45h	C01.44	Width level of the 2nd notch	-	0-4000	0	0.1%	U16	During operation	Immediately
46h	C01.45	Depth level of the 2nd notch	-	10-1000	1000	0.1%	U16	During operation	Immediately
47h	C01.46	Frequency of the 3rd notch	-	10-8000	8000	Hz	U16	During operation	Immediately
48h	C01.47	Width level of the 3rd notch	-	0-4000	0	0.1%	U16	During operation	Immediately
49h	C01.48	Depth level of the 3rd notch	-	10-1000	1000	0.1%	U16	During operation	Immediately
4Ah	C01.49	Frequency of the 4th notch	-	10-8000	8000	Hz	U16	During operation	Immediately
4Bh	C01.4A	Width level of the 4th notch	-	0-4000	0	0.1%	U16	During operation	Immediately
4Ch	C01.4B	Depth level of the 4th notch	-	10-1000	1000	0.1%	U16	During operation	Immediately
4Dh	C01.4C	Frequency of the 5th notch	-	10-8000	8000	Hz	U16	During operation	Immediately
4Eh	C01.4D	Width level of the 5th notch	-	0-4000	0	0.1%	U16	During operation	Immediately
4Fh	C01.4E	Depth level of the 5th notch	-	10-1000	1000	0.1%	U16	During operation	Immediately

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- For details about parameters above, refer to section 11.3.2 "Group C01".

Advanced Gain Parameters (2002h/C02)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	C02.00	Model tracking control	0: Disabled 1: Single mass model tracking	0-1	0	-	U16	At stop	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
02h	C02.01	Model tracking control gain	-	10-20000	500	0.1rad/s	U16	During operation	Immediately
03h	C02.02	Model tracking inertia correction coefficient	-	10-8000	1000	0.1%	U16	During operation	Immediately
31h	C02.30	Speed observer gain	-	0-40000	0	0.1Hz	U16	During operation	Immediately
32h	C02.31	Speed observer inertia correction	-	10-8000	1000	0.1%	U16	During operation	Immediately
33h	C02.32	Speed observer speed feedback cutoff frequency	-	0-16000	0	Hz	U16	During operation	Immediately
39h	C02.38	Frequency for vibration suppression 1	-	10-20000	1000	0.1Hz	U16	During operation	Immediately
3Ah	C02.39	Inertia correction for vibration suppression 1	-	10-8000	1000	0.1%	U16	During operation	Immediately
3Bh	C02.3A	Low-pass filter correction for vibration suppression 1	-	-9999-9999	0	0.1Hz	I16	During operation	Immediately
3Ch	C02.3B	Correction of high-pass filter 1 for vibration suppression 1	-	-9999-9999	0	0.1Hz	I16	During operation	Immediately
3Dh	C02.3C	Frequency of high-pass filter 2 for vibration suppression 1	-	10-50000	20000	0.1Hz	U16	During operation	Immediately
3Eh	C02.3D	Ratio of compensation 1 for vibration suppression 1	-	0-20000	0	0.1%	U16	During operation	Immediately
3Fh	C02.3E	Ratio of compensation 2 for vibration suppression 1	-	0-20000	0	0.1%	U16	During operation	Immediately
41h	C02.40	Frequency for vibration suppression 2	-	10-20000	1000	0.1Hz	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
42h	C02.41	Inertia correction for vibration suppression 2	-	10-8000	1000	0.1%	U16	During operation	Immediately
43h	C02.42	Low-pass filter correction for vibration suppression 2	-	-9999-9999	0	0.1Hz	I16	During operation	Immediately
44h	C02.43	Correction of high-pass filter 1 for vibration suppression 2	-	-9999-9999	0	0.1Hz	I16	During operation	Immediately
45h	C02.44	Frequency of high-pass filter 2 for vibration suppression 2	-	10-50000	20000	0.1Hz	U16	During operation	Immediately
46h	C02.45	Ratio of compensation 1 for vibration suppression 2	-	0-20000	0	0.1%	U16	During operation	Immediately
47h	C02.46	Ratio of compensation 2 for vibration suppression 2	-	0-20000	0	0.1%	U16	During operation	Immediately
49h	C02.48	Frequency for vibration suppression 3	-	10-20000	1000	0.1Hz	U16	During operation	Immediately
4Ah	C02.49	Inertia correction for vibration suppression 3	-	10-8000	1000	0.1%	U16	During operation	Immediately
4Bh	C02.4A	Low-pass filter correction for vibration suppression 3	-	-9999-9999	0	0.1Hz	I16	During operation	Immediately
4Ch	C02.4B	Correction of high-pass filter 1 for vibration suppression 3	-	-9999-9999	0	0.1Hz	I16	During operation	Immediately
4Dh	C02.4C	Frequency of high-pass filter 2 for vibration suppression 3	-	10-50000	20000	0.1Hz	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
4Eh	C02.4D	Ratio of compensation 1 for vibration suppression 3	-	0-20000	0	0.1%	U16	During operation	Immediately
4Fh	C02.4E	Ratio of compensation 2 for vibration suppression 3	-	0-20000	0	0.1%	U16	During operation	Immediately
61h	C02.60	Disturbance observer gain	-	0-40000	0	0.1Hz	U16	During operation	Immediately
62h	C02.61	Disturbance observer inertia correction coefficient	-	1-10000	1000	0.1%	U16	During operation	Immediately
63h	C02.62	Disturbance observer low-pass cutoff frequency	-	0-16000	0	Hz	U16	During operation	Immediately
64h	C02.63	Disturbance observer compensation torque percentage	-	0-2000	0	0.1%	U16	During operation	Immediately
69h	C02.68	Friction compensation switch and relevant setting	-	0-255	0	-	U16	During operation	Immediately
6Ah	C02.69	Friction compensation speed threshold	-	0-5000	20	0.1rpm	U16	During operation	Immediately
6Bh	C02.6A	Static friction compensation	-	0-2000	0	0.1%	U16	During operation	Immediately
6Ch	C02.6B	Forward friction compensation of coulomb friction	-	0-2000	0	0.1%	U16	During operation	Immediately
6Dh	C02.6C	Reverse friction compensation of coulomb friction	-	-2000-0	0	0.1%	I16	During operation	Immediately
6Eh	C02.6D	Viscous friction torque for rated speed	-	0-2000	0	0.1%	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
6Fh	C02.6E	Friction compensation filter time	-	0-65535	0	0.01ms	U16	During operation	Immediately
70h	C02.6F	Friction compensation threshold for zero speed	-	0-1000	10	0.1rpm	U16	During operation	Immediately

Instruction Parameters (2003h/C03)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
22h	C03.21	Speed reference	-	-8000-8000	100	rpm	I16	During operation	Immediately
23h	C03.22	Acceleration rate	-	0-3600000	10	ms	U32	During operation	Immediately
25h	C03.24	Deceleration rate	-	0-3600000	10	ms	U32	During operation	Immediately
28h	C03.27	Internal positive speed limit	-	0-8000	6000	rpm	U16	During operation	Immediately
29h	C03.28	Internal negative speed limit	-	0-8000	6000	rpm	U16	During operation	Immediately
2Ch	C03.2B	Speed reach threshold	-	0-8000	1000	rpm	U16	During operation	Immediately
2Dh	C03.2C	Speed synchronization threshold	-	0-1000	10	rpm	U16	During operation	Immediately
2Eh	C03.2D	Speed rotation threshold	-	0-1000	20	rpm	U16	During operation	Immediately
2Fh	C03.2E	Zero speed output threshold	-	0-1000	10	rpm	U16	During operation	Immediately
42h	C03.41	Torque reference	-	-4000-4000	0	0.1%	I16	During operation	Immediately
44h	C03.43	Internal positive torque limit	-	0-4000	3000	0.1%	U16	During operation	Immediately
45h	C03.44	Internal negative torque limit	-	0-4000	3000	0.1%	U16	During operation	Immediately
48h	C03.47	Positive speed limit in torque mode	-	0-8000	3000	rpm	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
49h	C03.48	Negative speed limit in torque mode	-	0-8000	3000	rpm	U16	During operation	Immediately
4Ah	C03.49	Reference value for torque reach	-	0-4000	0	0.1%	U16	During operation	Immediately
4Bh	C03.4A	Valid value for torque reached	-	0-4000	200	0.1%	U16	During operation	Immediately
4Ch	C03.4B	Invalid value for torque reached	-	0-4000	100	0.1%	U16	During operation	Immediately

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- For details about parameters above, refer to section 11.3.3 "Group C03".

I/O Parameters (2004h/C04)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	C04.00	DI1 function selection	0: No definition 1: S-ON 2: Fault reset 4: Emergency stop 5: Home switch 6: Forward overtravel 7: Reverse overtravel 30: Probe 1 31: Probe 2	0-32	6	-	U16	At stop	Immediately
02h	C04.01	DI1 logic selection	0: Active low 1: Active high	0-1	0	-	U16	During operation	Immediately
03h	C04.02	DI1 filter time	-	0-65535	150	0.01ms	U16	During operation	Immediately
05h	C04.04	DI2 function selection	0: No definition 1: S-ON 2: Fault reset 4: Emergency stop 5: Home switch 6: Forward overtravel 7: Reverse overtravel 30: Probe 1 31: Probe 2	0-32	7	-	U16	At stop	Immediately
06h	C04.05	DI2 logic selection	0: Active low 1: Active high	0-1	0	-	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
07h	C04.06	DI2 filter time	-	0-65535	150	0.01ms	U16	During operation	Immediately
09h	C04.08	DI3 function selection	0: No definition 1: S-ON 2: Fault reset 4: Emergency stop 5: Home switch 6: Forward overtravel 7: Reverse overtravel 30: Probe 1 31: Probe 2	0-32	5	-	U16	At stop	Immediately
0Ah	C04.09	DI3 logic selection	0: Active low 1: Active high	0-1	0	-	U16	During operation	Immediately
0Bh	C04.0A	DI3 filter time	-	0-65535	150	0.01ms	U16	During operation	Immediately
0Dh	C04.0C	DI4 function selection	0: No definition 1: S-ON 2: Fault reset 4: Emergency stop 5: Home switch 6: Forward overtravel 7: Reverse overtravel 30: Probe 1 31: Probe 2	0-32	31	-	U16	At stop	Immediately
0Eh	C04.0D	DI4 logic selection	0: Active low 1: Active high	0-1	0	-	U16	During operation	Immediately
0Fh	C04.0E	DI4 filter time	-	0-65535	150	0.01ms	U16	During operation	Immediately
11h	C04.10	DI5 function selection	0: No definition 1: S-ON 2: Fault reset 4: Emergency stop 5: Home switch 6: Forward overtravel 7: Reverse overtravel 30: Probe 1 31: Probe 2	0-32	30	-	U16	At stop	Immediately
12h	C04.11	DI5 logic selection	0: Active low 1: Active high	0-1	0	-	U16	During operation	Immediately
13h	C04.12	DI5 filter time	-	0-65535	150	0.01ms	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
31h	C04.30	DO1 function selection	0: No definition 1: Servo ready 2: Motor rotation 9: Brake output 10: Alarm 11: Fault 32: EDM safety state	0-20	1	-	U16	At stop	Immediately
32h	C04.31	DO1 logic selection	0: Active low 1: Active high	0-1	0	-	U16	During operation	Immediately
33h	C04.32	DO2 function selection	0: No definition 1: Servo ready 2: Motor rotation 9: Brake output 10: Alarm 11: Fault 32: EDM safety state	0-20	4	-	U16	At stop	Immediately
34h	C04.33	DO2 logic selection	0: Active low 1: Active high	0-1	0	-	U16	During operation	Immediately
35h	C04.34	DO3 function selection	0: No definition 1: Servo ready 2: Motor rotation 9: Brake output 10: Alarm 11: Fault 32: EDM safety state	0-20	3	-	U16	At stop	Immediately
36h	C04.35	DO3 logic selection	0: Active low 1: Active high	0-1	0	-	U16	During operation	Immediately

Stop Mode (2005h/C05)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
03h	C05.02	Stop mode at overtravel	0: Coast to stop, keeping de-energized status 1: Stop at zero speed, keeping position lock status 2: Stop at zero speed, keeping de-energized status 3: Ramp to stop as defined by 6085h, keeping de-energized status 4: Ramp to stop as defined by 6085h, keeping position lock status 5: Dynamic braking stop, keeping de-energized status 6: Dynamic braking stop, keeping dynamic braking status 7: Not responding to overtravel	0-7	1	-	U16	At stop	Immediately
04h	C05.03	Stop mode at No. 1 fault	0: Coast to stop, keeping de-energized status 1: Dynamic braking stop, keeping de-energized status 2: Dynamic braking stop, keeping dynamic braking status	0-2	2	-	U16	At stop	Immediately
0Dh	C05.0C	Limit for stop at emergency-stop torque	-	0-3000	1000	0.1%	U16	During operation	Immediately
0Eh	C05.0D	Maximum downtime	-	0-65535	10000	ms	U16	At stop	Immediately
11h	C05.10	Delay from brake close to motor de-energized	-	0-65535	100	ms	U16	During operation	Immediately
12h	C05.11	Speed threshold at brake closing	-	10-3000	30	rpm	U16	During operation	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
13h	C05.12	Maximum waiting time with S-ON off at brake closing	-	0-65535	100	ms	U16	During operation	Immediately
14h	C05.13	Delay from brake on to command received	-	0-65535	100	ms	U16	At stop	Immediately
15h	C05.14	Energizing delay of DB relay	-	0-65535	20	ms	U16	At stop	Immediately

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- For details about parameters above, refer to section 11.3.4 "Group C05".

Protection Parameters (2006h/C06)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
04h	C06.03	Threshold of excessive speed	-	0-9000	0	rpm	U16	During operation	Immediately
05h	C06.04	Input phase loss detection	0: Enabled 1: Disabled	0-1	0	-	U16	At stop	Immediately
06h	C06.05	Retentive at power failure	0: Non-retentive 1: Retentive	0-1	0	-	U16	At stop	Immediately
08h	C06.07	Mechanical limit position	0: Inactive 1: Enabled 2: Enabled after homing	0-2	0	-	U16	During operation	Immediately
09h	C06.08	Mechanical PL	-	-2^{31} ($2^{31}-1$)	$2^{31}-1$	Unit in application	I32	During operation	Immediately
0Bh	C06.0A	Mechanical NL	-	-2^{31} ($2^{31}-1$)	-2^{31}	Unit in application	I32	During operation	Immediately
11h	C06.10	Drive overload protection threshold	-	0-3500	1150	0.1%	U16	At stop	Immediately
12h	C06.11	Motor overload protection threshold	-	0-3500	1150	0.1%	U16	At stop	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
13h	C06.12	Motor locked-rotor detection	0: Inactive 1: Enabled	0-1	1	-	U16	At stop	Immediately
14h	C06.13	Motor locked-rotor detection time	-	0-3000	200	ms	U16	At stop	Immediately
15h	C06.14	Motor locked-rotor detection speed	-	0-1000	10	rpm	U16	During operation	Immediately
16h	C06.15	Output phase loss detection	0: Inactive 1: Enabled	0-1	0	-	U16	At stop	Immediately
1Dh	C06.1C	Encoder communication fault tolerance threshold	-	0-88	3	-	U16	At stop	Immediately
21h	C06.20	Protection from out of control	0: Inactive 1: Enabled	0-1	1	-	U16	At stop	Immediately

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- For details about parameters above, refer to section 11.3.5 "Group C06".

Auto-tuning Parameters (2007h/C07)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	C07.00	Offline inertia auto-tuning mode setting	-	0-785	769	-	U16	At stop	Immediately
02h	C07.01	Offline inertia auto-tuning speed reference	-	50-1000	500	rpm	U16	At stop	Immediately
03h	C07.02	Acceleration/Deceleration time for offline inertia auto-tuning	-	0-65535	100	ms	U16	At stop	Immediately
04h	C07.03	Offline inertia auto-tuning target torque	-	1-1500	150	0.1%	U16	At stop	Immediately
05h	C07.04	Offline inertia auto-tuning revolutions	-	10-65535	200	0.01r	U16	At stop	Immediately

Communication Parameters (200Ah/COA)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
09h	COA.08	Commissioning software communication station ID	-	1-255	1	-	U16	At stop	Immediately
0Ah	COA.09	Commissioning software communication baud rate	0: 1200bps 1: 2400bps 2: 4800bps 3: 9600bps 4: 19200bps 5: 38400bps 6: 57600bps 7: 115200bps	0-7	7	-	U16	At stop	Upon re-power-on
0Bh	COA.0A	Commissioning software communication format	0: No parity, 1 stop bit 1: Odd parity, 1 stop bit 2: Even parity, 1 stop bit 3: No parity, 2 stop bits 4: Odd parity, 2 stop bits 5: Even parity, 2 stop bits	0-5	0	-	U16	At stop	Upon re-power-on
0Ch	COA.0B	Commissioning software communication response time	-	1-1000	1	ms	U16	At stop	Immediately
0Dh	COA.0C	Commissioning software communication timeout	-	0-65535	0	-	U16	Read only	Immediately
0Eh	COA.0D	Commissioning software communication storage	0: No storage 1: Storage	0-1	1	-	U16	At stop	Immediately
0Fh	COA.0E	Commissioning software data format	0: Low 16 bits before high 16 bits 1: High 16 bits before low 16 bits	0-1	0	-	U16	At stop	Immediately

NOTICE

- For details about parameters above, refer to section 11.3.6 "Group COA".

Homing Touch Probe Parameters (2010h/C10)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	C10.00	Homing enable	0: Inactive 1: Written through communication 2: DI trigger 3: Current position as home	0-3	0	-	U16	During operation	Immediately
09h	C10.08	Homing timeout interval	-	$0-(2^{32}-1)$	60000	ms	U32	During operation	Immediately
11h	C10.10	Multi-turn absolute position offset (low 32 bits)	-	$-2^{31}-(2^{31}-1)$	0	P	I32	At stop	Upon re-power-on
13h	C10.12	Multi-turn absolute position offset (high 32 bits)	-	$-2^{31}-(2^{31}-1)$	0	P	I32	At stop	Upon re-power-on
15h	C10.14	Multi-turn revolutions data offset	-	0-65535	0	Rev	U16	Read only	Immediately
16h	C10.15	Multi-turn overflow flag	-	0-1	0	-	U16	Read only	Immediately
17h	C10.16	Reference running mode in rotation mode	0: Nearest 1: Always in forward direction 2: Always in reverse direction 3: Always in current direction 4: Not specified	0-4	0	-	U16	At stop	Immediately
19h	C10.18	Numerator of electronic gear ratio in rotation mode	-	1-65535	1	-	U16	At stop	Immediately
1Ah	C10.19	Denominator of electronic gear ratio in rotation mode	-	1-65535	1	-	U16	At stop	Immediately
1Bh	C10.1A	Upper limit of mechanical absolute position in rotation mode (low 32 bits)	-	$0-(2^{32}-1)$	0	P	U32	At stop	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
1Dh	C10.1C	Upper limit of mechanical absolute position in ro-tation mode (high 32 bits)	-	$0-(2^{32}-1)$	0	P	U32	At stop	Immediately
1Fh	C10.1E	Single-turn homing absolute value offset	-	$-2^{31}-(2^{31}-1)$	0	Unit in application	I32	At stop	Upon re-power-on
31h	C10.30	Torque limit of homing upon hit-and-stop	-	0-3000	1000	0.1%	U16	During operation	Immediately
32h	C10.31	Speed for homing upon hit-and-stop	-	0-1000	10	rpm	U16	During operation	Immediately
33h	C10.32	Number of times for homing upon hit-and-stop	-	0-65535	30	-	U16	During operation	Immediately

EtherCAT Parameters (2013h/C13)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	C13.00	EtherCAT slave name	-	0-65535	0	-	U16	Read only	Immediately
02h	C13.01	EtherCAT slave alias	-	0-65535	0	-	U16	During operation	Immediately
03h	C13.02	EtherCAT sync loss threshold	-	1-20	8	-	U16	During operation	Immediately
04h	C13.03	EtherCAT synchronization detection mode	-	0-65535	0	-	U16	At stop	Immediately
05h	C13.04	EtherCAT sync loss count	-	0-65535	0	-	U16	Read only	Immediately
06h	C13.05	EtherCAT synchronization mode setting	-	0-2	1	-	U16	During operation	Immediately
07h	C13.06	EtherCAT synchronization error threshold	-	0-6000	3000	ns	U16	At stop	Immediately
08h	C13.07	Occurrence count of excessive position reference increment in sync position mode	-	1-30	5	-	U16	During operation	Immediately
09h	C13.08	EtherCAT enhanced link selection	0: Inactive 1: Enabled	0-1	0	-	U16	During operation	Upon re-power-on

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
0Ah	C13.09	Maximum errors and invalid frames of EtherCAT port 0 per unit time	-	0-65535	0	-	U16	Read only	Immediately
0Bh	C13.0A	Maximum errors and invalid frames of EtherCAT port 1 per unit time	-	0-65535	0	-	U16	Read only	Immediately
0Ch	C13.0B	Max. transfer error of EtherCAT port per unit time	-	0-65535	0	-	U16	Read only	Immediately
0Dh	C13.0C	Max. EtherCAT data frame processing unit error per unit time	-	0-65535	0	-	U16	Read only	Immediately
0Eh	C13.0D	Max. link loss value of EtherCAT port per unit time	-	0-65535	0	-	U16	Read only	Immediately
0Fh	C13.0E	EtherCAT state machine status and port connection status	-	0-65535	0	-	U16	Read only	Immediately
10h	C13.0F	EtherCAT AL status code	-	0-65535	0	-	U16	Read only	Immediately
11h	C13.10	EtherCAT parameter storage	0: No storage 1: Storage	0-1	1	-	U16	During operation	Immediately
12h	C13.11	EtherCAT IRQ loss threshold	-	0-10	5	-	U16	During operation	Immediately
13h	C13.12	EtherCAT IRQ loss count	-	0-65535	0	-	U16	Read only	Immediately
1Ah	C13.19	Use of the loop network	-	0-1	0	-	U16	At stop	Immediately

NOTICE

- For details about parameters above, refer to section 11.3.7 "Group C13".

Motor Parameters (2020h/R20)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	R20.00	Motor model	-	0-65535	20000	-	U16	At stop	Upon re-power-on

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
23h	R20.22	Encoder type	-	0-65535	0	-	U16	Read only	Upon re-power-on

Drive Parameters (2021h/R21)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	R21.00	Drive model	-	0-65535	3	-	U16	At stop	Upon re-power-on
02h	R21.01	Internal drive model	-	0-65535	3	-	U16	Read only	Immediately
0Dh	R21.0C	Drive voltage class	-	0-2	0	-	U16	Read only	Immediately
0Eh	R21.0D	Rated drive power	-	$1-(2^{32}-1)$	40	0.01kW	U32	Read only	Immediately
10h	R21.0F	Rated output current of drive	-	$1-(2^{32}-1)$	280	0.01A	U32	Read only	Immediately
12h	R21.11	Maximum output current of drive	-	$1-(2^{32}-1)$	980	0.01A	U32	Read only	Immediately
14h	R21.13	Internal bleeder resistor power	-	1-65535	40	W	U16	Read only	Immediately
15h	R21.14	Internal bleeder resistor resistance	-	1-65535	50	Ω	U16	Read only	Immediately

NOTICE

- For details about parameters above, refer to section 11.3.8 "Group R21".

Motor Gain Parameters (2022h/R22)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	R22.00	Current loop mode	0: Standard mode 1: Performance mode	0-1	0	-	U16	At stop	Upon re-power-on
02h	R22.01	Current loop response level	-	0-4000	0	0.1%	U16	At stop	Upon re-power-on
21h	R22.20	MTPA field-weakening switch	-	0-65535	256	-	U16	At stop	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
22h	R22.21	Field-weakening depth	-	500-2000	1000	0.1%	U16	At stop	Immediately
23h	R22.22	Field-weakening proportional gain	-	10-1000	100	Hz	U16	At stop	Immediately
24h	R22.23	Field-weakening integral gain	-	0-8000	100	0.1%	U16	At stop	Immediately
25h	R22.24	Cutoff frequency of d axis current low-pass filter	-	0-16000	0	Hz	U16	At stop	Immediately
26h	R22.25	Field-weakening d axis current limit	-	0-3000	1500	0.1%	U16	At stop	Immediately
31h	R22.30	Dead zone compensation	-	0-2000	1000	0.1%	U16	At stop	Immediately

Parameters of Control in Progress (2030h/F30)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	F30.00	JOG enabling in velocity mode	-	0~8000	0	-	U16	During operation	Immediately
02h	F30.01	JOG enabling in position mode	-	0~8000	0	-	U16	During operation	Immediately
03h	F30.02	JOG velocity reference	-	0~8000	100	rpm	U16	During operation	Immediately
04h	F30.03	JOG acceleration/ deceleration time	-	0~3600000	100	ms	U32	During operation	Immediately
06h	F30.05	JOG distance in position mode	-	$-2^{31} - (2^{31} - 1)$	20000	Unit in application	I32	During operation	Immediately
11h	F30.10	Inertia auto-tuning selection	0: Disabled 1: Enabled	0-65535	0	-	U16	During operation	Immediately
12h	F30.11	Initial angle auto-tuning selection	0: Disabled 1: Enabled	0-1	0	-	U16	During operation	Immediately

NOTICE

- For details about parameters above, refer to section 11.3.9 "Group F30".

Parameters of Control in Progress (2031h/F31)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	F31.00	Fault reset	0: Inactive 1: Reset	0-1	0	-	U16	At stop	Immediately
02h	F31.01	Software reset	0: Inactive 1: Reset	0-1	0	-	U16	At stop	Immediately
03h	F31.02	Parameter initialization	0: Inactive 1: Restore default settings of parameters 2: Restore default settings of the object dictionary	0-2	0	-	U16	At stop	Immediately
04h	F31.03	Drive motor parameter reset	0: Inactive 1: Factory reset drive parameters 2: Factory reset motor parameters	0-2	0	-	U16	At stop	Immediately
05h	F31.04	Fault record initialization	0: Inactive 1: Fault record clearing	0-1	0	-	U16	At stop	Immediately
11h	F31.10	Encoder data reset	0: Inactive 1: Read encoder 2: Write encoder 3: Reset encoder fault 4: Reset encoder fault and multi-turn data 16: Operation failed	0-31	0	-	U16	At stop	Immediately

NOTICE

- For details about parameters above, refer to section 11.3.10 "Group F31".

Running Monitoring Parameters (2040h/U40)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	U40.00	Speed reference	-	-9000-9000	0	rpm	I16	Read only	Immediately
02h	U40.01	Speed feedback	-	-9000-9000	0	rpm	I16	Read only	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
03h	U40.02	Torque reference	-	-4000-4000	0	0.1%	I16	Read only	Immediately
04h	U40.03	Torque feedback	-	-4000-4000	0	0.1%	I16	Read only	Immediately
05h	U40.04	DI status	-	0-65535	0	-	U16	Read only	Immediately
06h	U40.05	DO status	-	0-65535	0	-	U16	Read only	Immediately
07h	U40.06	Bus voltage	-	0-9000	0	0.1V	U16	Read only	Immediately
08h	U40.07	Average load ratio	-	0-4000	0	0.1%	U16	Read only	Immediately
09h	U40.08	Electrical angle	-	0-36000	0	0.01°	U16	Read only	Immediately
0Ah	U40.09	Mechanical angle	-	0-36000	0	0.01°	U16	Read only	Immediately
0Dh	U40.0C	RMS value of phase current	-	-9000-9000	0	0.1A	I16	Read only	Immediately
11h	U40.10	Position deviation counter	-	$-2^{31} - (2^{31}-1)$	0	P	I32	Read only	Immediately
15h	U40.14	Absolute position reference	-	$-2^{31} - (2^{31}-1)$	0	Unit in application	I32	Read only	Immediately
17h	U40.16	Absolute position feedback (reference unit)	-	$-2^{31} - (2^{31}-1)$	0	Unit in application	I32	Read only	Immediately
19h	U40.18	Absolute position feedback (encoder unit)	-	$-2^{31} - (2^{31}-1)$	0	P	I32	Read only	Immediately
1Bh	U40.1A	Absolute position feedback (encoder unit)	-	$-2^{31} - (2^{31}-1)$	0	P	I32	Read only	Immediately
1Dh	U40.1C	Encoder single-turn data	-	$-2^{31} - (2^{31}-1)$	0	P	I32	Read only	Immediately
1Fh	U40.1E	Encoder multi-turn position data	-	0-65535	0	Rev	U16	Read only	Immediately
20h	U40.1F	Encoder initial angle	-	0-36000	0	0.01°	U16	Read only	Immediately
21h	U40.20	Encoder multi-turn data (low 32 bits)	-	$-2^{31} - (2^{31}-1)$	0	P	I32	Read only	Immediately
23h	U40.22	Encoder multi-turn data (high 32 bits)	-	$-2^{31} - (2^{31}-1)$	0	P	I32	Read only	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
25h	U40.24	Absolute position feedback (encoder unit) (low 32 bits)	-	$-2^{31}-1$	0	P	I32	Read only	Immediately
27h	U40.26	Absolute position feedback (encoder unit) (high 32 bits)	-	$-2^{31}-1$	0	P	I32	Read only	Immediately
29h	U40.28	Position feedback in rotation mode (reference unit) (low 32 bits)	-	$-2^{31}-1$	0	Unit in application	I32	Read only	Immediately
2Bh	U40.2A	Position feedback in rotation mode (encoder unit) (low 32 bits)	-	$-2^{31}-1$	0	P	I32	Read only	Immediately
2Dh	U40.2C	Position feedback in rotation mode (encoder unit) (high 32 bits)	-	$-2^{31}-1$	0	P	I32	Read only	Immediately
31h	U40.30	Heatsink temperature	-	-9000-9000	0	0.1°C	I16	Read only	Immediately
35h	U40.34	Offline inertia auto-tuning value	-	0-12000	0	%	U16	Read only	Immediately
37h	U40.36	Instantaneous value in phase U current	-	$-2^{31}-1$	0	0.001A	I32	Read only	Immediately
39h	U40.38	Instantaneous value in phase V current	-	$-2^{31}-1$	0	0.001A	I32	Read only	Immediately
3Bh	U40.3A	Synchronization cycle measured value	-	$0-(2^{31}-1)$	0	10ns	U32	Read only	Immediately
3Dh	U40.3C	SYNC and IRQ phase value	-	$-2^{31}-1$	0	10ns	I32	Read only	Immediately
3Fh	U40.3E	Drive accumulated heat	-	0-2000	0	0.1%	U16	Read only	Immediately
40h	U40.3F	Motor accumulated heat	-	0-2000	0	0.1%	U16	Read only	Immediately

NOTICE

- For details about parameters above, refer to section 11.3.11 "Group U40".

Status Monitoring Parameters (2041h/U41)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	U41.00	MCU system status	-	0-65535	0	-	U16	Read only	Immediately
02h	U41.01	MCU fault state	-	0-65535	0	-	U16	Read only	Immediately
05h	U41.04	Encoder system status	-	0-65535	0	-	U16	Read only	Immediately
06h	U41.05	Encoder fault state	-	0-65535	0	-	U16	Read only	Immediately
07h	U41.06	Group number of abnormal parameter	-	0-255	0	-	U16	Read only	Immediately
08h	U41.07	Offset of the abnormal parameter within the parameter group	-	0-255	0	-	U16	Read only	Immediately
0Bh	U41.0A	Servo Status	-	0-3	0	-	U16	Read only	Immediately
0Ch	U41.0B	Servo running mode	-	0-9	0	-	U16	Read only	Immediately
0Dh	U41.0C	Servo running time	-	$0-(2^{32}-1)$	0	0.1s	U32	Read only	Immediately

Version Parameters (2042h/U42)

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
01h	U42.00	ARM version	-	0-65535	0	0.01	U16	Read only	Immediately
03h	U42.02	Encoder version	-	0-65535	0	0.01	U16	Read only	Immediately
04h	U42.03	ARM-based machine	-	0-65535	0	0.01	U16	Read only	Immediately
06h	U42.05	Internal software version	-	0-65535	0	0.01	U16	Read only	Immediately
0Bh	U42.0A	EtherCAT CoE version	-	0-65535	0	0.01	U16	Read only	Immediately
0Ch	U42.0B	EtherCAT XML version	-	0-65535	0	0.01	U16	Read only	Immediately
11h	U42.10	Drive model	-	0-65535	0	-	U16	Read only	Immediately
12h	U42.11	Motor model	-	0-65535	0	-	U16	Read only	Immediately
13h	U42.12	Encoder model	-	0-65535	0	-	U16	Read only	Immediately
14h	U42.13	Power supply unit model identification	-	0-65535	0	-	U16	Read only	Immediately

Index	Parameter	Name	Options	Value Range	Default	Unit	Data Type	Modification Mode	Effective Time
15h	U42.14	Inverter model identification 1	-	0-65535	0	-	U16	Read only	Immediately
16h	U42.15	Inverter model identification 2	-	0-65535	0	-	U16	Read only	Immediately
17h	U42.16	Servo version	-	0-65535	0	-	U16	Read only	Immediately

11.2.2 Common Parameters in Group 6000h

Parameter group 6000h contains supported sub-protocol DSP 402 related objects.

Index	Sub-index	Name	Access the Platform	PDO Mapping	Data Type	Unit	Value Range	Default	Modification Mode	Effective Time
603Fh	0	Error code	RO	TPDO	U16	-	-	-	-	-
6040h	0	Control word	RW	RPDO	U16	-	0-65535	0	During operation	Immediately
6041h	0	Status word	RO	TPDO	U16	-	-	-	-	-
605Ah	0	Quick stop option code	RW	NO	I16	-	0-7	2	During operation	Upon re-power-on
605Ch	0	Stop mode upon servo-off	RW	NO	I16	-	-4-1	0	During operation	Upon re-power-on
605Dh	0	Halt option code	RW	NO	I16	-	1-3	1	During operation	Upon re-power-on
605Eh	0	Stop mode at No. 2 fault	RW	NO	I16	-	5-3	2	During operation	Upon re-power-on
6060h	0	Servo mode	RW	RPDO	I8	-	0-10	0	During operation	Immediately
6061h	0	Modes of operation display	RO	TPDO	I8	-	-	-	-	-
6062h	0	Position reference	RO	TPDO	I32	Reference unit	-	-	-	-
6063h	0	Position feedback	RO	TPDO	I32	Encoder unit	-	-	-	-
6064h	0	Position feedback	RO	TPDO	I32	Reference unit	-	-	-	-
6065h	0	Following error window	RW	RPDO	U32	Reference unit	$0-(2^{32}-1)$	0	During operation	Immediately
6066h	0	Following error time out	RW	RPDO	U16	ms	0-65535	0	During operation	Immediately
6067h	0	Position reach threshold	RW	RPDO	U32	Reference unit	$0-(2^{32}-1)$	734	During operation	Immediately

Index	Sub-index	Name	Access the Platform	PDO Mapping	Data Type	Unit	Value Range	Default	Modification Mode	Effective Time
6068h	0	Position window time	RW	RPDO	U16	ms	0-65535	0	During operation	Immediately
606Ch	0	Actual speed	RO	TPDO	I32	Reference unit/s	-	-	-	-
606Dh	0	Speed reach threshold	RW	RPDO	U16	rpm	0-65535	10	During operation	Immediately
606Eh	0	Velocity window time	RW	RPDO	U16	ms	0-65535	0	During operation	Immediately
606Fh	0	Velocity threshold	RW	RPDO	U16	rpm	0-65535	10	During operation	Immediately
6070h	0	Velocity threshold time	RW	RPDO	U16	ms	0-65535	0	During operation	Immediately
6071h	0	Target torque	RW	RPDO	I16	0.1%	4000-4000	0	During operation	Immediately
6072h	0	Max. torque	RW	RPDO	U16	0.1%	0-4000	3500	During operation	Immediately
6074h	0	Torque reference	RO	TPDO	I16	0.1%	-	0	-	-
6077h	0	Actual torque	RO	TPDO	I16	0.1%	-	0	-	-
607Ah	0	Target position	RW	RPDO	I32	Reference unit	$-2^{31}-(2^{31}-1)$	0	During operation	Immediately
607Ch	0	Home offset	RW	RPDO	I32	Reference unit	$-2^{31}-(2^{31}-1)$	0	During operation	Immediately
607D	Software absolute position limit									
	0	Highest sub-index supported	RO	NO	U8	-	-	0x02	-	-
	1	Minimum software position limit	RW	RPDO	I32	Reference unit	$-2^{31}-(2^{31}-1)$	-2^{31}	During operation	Immediately
	2	Maximum software position limit	RW	RPDO	I32	Reference unit	$-2^{31}-(2^{31}-1)$	$2^{31}-1$	During operation	Immediately
607Eh	0	Reference polarity	RW	RPDO	U8	-	0-255	0	During operation	Immediately
607Fh	0	Maximum speed	RW	RPDO	U32	Reference unit/s	$0-(2^{32}-1)$	104857600	During operation	Immediately
6081h	0	Profile operating speed	RW	RPDO	U32	User velocity	$0-(2^{32}-1)$	1747627	During operation	Immediately
6083h	0	Profile acceleration rate	RW	RPDO	U32	Reference unit/s ²	$0-(2^{32}-1)$	174762666	During operation	Immediately

Index	Sub-index	Name	Access the Platform	PDO Mapping	Data Type	Unit	Value Range	Default	Modification Mode	Effective Time
6084h	0	Profile deceleration rate	RW	RPDO	U32	Reference unit/s ²	0-(2 ³² -1)	174762666	During operation	Immediately
6085h	0	Quick stop deceleration	RW	RPDO	U32	Reference unit/s ²	0-(2 ³² -1)	2 ³¹ -1	During operation	Immediately
6086h	0	Motion profile type	RW	RPDO	I16	-	32767-32767	0	During operation	Immediately
6087h	0	Torque slope	RW	RPDO	U32	0.1%/s	0-(2 ³² -1)	2 ³² -1	During operation	Immediately
6091h	Gear ratio									
	0	Highest sub-index supported	RO	NO	U8	U8	-	0x02	-	-
	1	Motor revolutions	RW	RPDO	U32	-	0-(2 ³² -1)	1	During operation	Immediately
	2	Shaft revolutions	RW	RPDO	U32	-	1-(2 ³² -1)	1	During operation	Immediately
6098h	0	Homing method	RW	RPDO	I8	-	2-35	1	During operation	Immediately
6099h	Homing speed									
	0	Highest sub-index supported	RO	NO	U8	-	-	2	-	-
	1	Speed during search for switch	RW	RPDO	U32	Reference unit/s	0-(2 ³² -1)	1747627	During operation	Immediately
	2	Speed during search for zero	RW	RPDO	U32	Reference unit/s	10-(2 ³² -1)	174763	During operation	Immediately
609Ah	0	Homing acceleration	RW	RPDO	U32	Reference unit/s ²	0-(2 ³² -1)	174762667	During operation	Immediately
60B0h	0	Position offset	RW	RPDO	I32	Reference unit	-2 ³¹ -(2 ³¹ -1)	0	During operation	Immediately
60B1h	0	Speed deviation	RW	RPDO	I32	Reference unit/s	-2 ³¹ -(2 ³¹ -1)	0	During operation	Immediately
60B2h	0	Torque offset	RW	RPDO	I16	0.10%	4000-4000	0	During operation	Immediately
60B8h	0	Touch probe function	RW	RPDO	U16	-	0-65535	0	During operation	Immediately
60B9h	0	Touch probe status	RW	TPDO	U16	-	-	0	-	-
60BAh	0	Touch probe 1 positive edge	RW	TPDO	I32	Reference unit	-	0	-	-

Index	Sub-index	Name	Access the Platform	PDO Mapping	Data Type	Unit	Value Range	Default	Modification Mode	Effective Time
60BBh	0	Touch probe 1 negative edge	RW	TPDO	I32	Reference unit	-	0	-	-
60BCh	0	Touch probe 2 positive edge	RW	TPDO	I32	Reference unit	-	0	-	-
60BDh	0	Touch probe 2 negative edge	RW	TPDO	I32	Reference unit	-	0	-	-
60C5h	0	Max. acceleration	RW	RPDO	U32	User acceleration unit	$0-2^{32}-1$	$2^{31}-1$	During operation	Immediately
60C6h	0	Max. deceleration	RW	RPDO	U32	User acceleration unit	$0-2^{32}-1$	$2^{31}-1$	During operation	Immediately
60D5h	0	Touch probe 1 positive edge counter	RO	TPDO	U16	-	-	0	-	-
60D6h	0	Touch probe 1 negative edge counter	RO	TPDO	U16	-	-	0	-	-
60D7h	0	Touch probe 2 positive edge counter	RO	TPDO	U16	-	-	0	-	-
60D8h	0	Touch probe 2 negative edge counter	RO	TPDO	U16	-	-	0	-	-
60E0h	0	Positive torque limit	RW	RPDO	U16	0.1%	0-4000	3500	During operation	Immediately
60E1h	0	Negative torque limit	RW	RPDO	U16	0.1%	0-4000	3500	During operation	Immediately
60E3h	Supported homing method									
	0	Highest sub-index supported	RO	NO	U8	-	-	22	-	-
	1	1st supported homing method	RO	NO	I16	-	-	1	-	-
	2	2nd supported homing method	RO	NO	I16	-	-	2	-	-
	3	3rd supported homing method	RO	NO	I16	-	-	3	-	-
	4	4th supported homing method	RO	NO	I16	-	-	4	-	-
	5	5th supported homing method	RO	NO	I16	-	-	5	-	-
	6	6th supported homing method	RO	NO	I16	-	-	6	-	-

Index	Sub-index	Name	Access the Platform	PDO Mapping	Data Type	Unit	Value Range	Default	Modification Mode	Effective Time
	7	7th supported homing method	RO	NO	I16	-	-	7	-	-
	8	8th supported homing method	RO	NO	I16	-	-	8	-	-
	9	9th supported homing method	RO	NO	I16	-	-	9	-	-
	A	10th supported homing method	RO	NO	I16	-	-	10	-	-
	B	11th supported homing method	RO	NO	I16	-	-	11	-	-
	C	12th supported homing method	RO	NO	I16	-	-	12	-	-
	D	13th supported homing method	RO	NO	I16	-	-	13	-	-
	E	14th supported homing method	RO	NO	I16	-	-	14	-	-
	F	15th supported homing method	RO	NO	I16	-	-	17	-	-
	10	16th supported homing method	RO	NO	I16	-	-	18	-	-
	11	17th supported homing method	RO	NO	I16	-	-	19	-	-
	12	18th supported homing method	RO	NO	I16	-	-	20	-	-
	13	19th supported homing method	RO	NO	I16	-	-	21	-	-
	14	20th supported homing method	RO	NO	I16	-	-	22	-	-
	15	21th supported homing method	RO	NO	I16	-	-	23	-	-
	16	22th supported homing method	RO	NO	I16	-	-	24	-	-
	17	23th supported homing method	RO	NO	I16	-	-	25	-	-
	18	24th supported homing method	RO	NO	I16	-	-	26	-	-
	19	25th supported homing method	RO	NO	I16	-	-	27	-	-

Index	Sub-index	Name	Access the Platform	PDO Mapping	Data Type	Unit	Value Range	Default	Modification Mode	Effective Time
	1A	26th supported homing method	RO	NO	I16	-	-	28	-	-
	1B	27th supported homing method	RO	NO	I16	-	-	29	-	-
	1C	28th supported homing method	RO	NO	I16	-	-	30	-	-
	1D	29th supported homing method	RO	NO	I16	-	-	33	-	-
	1E	30th supported homing method	RO	NO	I16	-	-	34	-	-
	1F	31th supported homing method	RO	NO	I16	-	-	35	-	-
60E6h	0	Encoder increments for the additional position	RW	NO	U16	-	0-1	0	During operation	Immediately
60F4h	0	Position deviation	RO	TPDO	I32	Reference unit	-	-	-	-
60FCh	0	Position reference	RO	TPDO	I32	Encoder unit	-	-	-	-
60FDh	0	DI status	RO	TPDO	U32	-	-	-	-	-
60FFh	0	Target velocity	RW	RPDO	I32	Reference unit/s	$(-2^{31}-1)-(2^{31}-1)$	0	During operation	Immediately
6502h	0	Supported drive modes	RO	NO	U32	-	-	941	-	-

NOTICE

- For details about parameters above, refer to section 11.3.12 "Group 6000".

11.3 Description of Parameters

11.3.1 Group C00

C00.05: Stiffness level

- Defines the stiffness level of the servo system. The higher the stiffness level, the stronger the gains and the quicker the response will be. But an excessively high stiffness level will cause vibration. The setpoint 0 indicates the lowest stiffness and 41 indicates the highest stiffness.

C00.06: Load inertia ratio

- Defines the mechanical load inertia ratio relative to the motor moment of inertia.
- When C00.06 is set to 0, it indicates the motor carries no load; if it is set to 1.00, it indicates the mechanical load inertia is the same as the motor moment of inertia.
- When the value of C00.06 is equal to the actual inertia ratio, the value of speed loop gain can represent the maximum follow-up frequency of actual speed loop.

11.3.2 Group C01

C01.00: 1st position loop gain

- Defines the proportional gain of the position loop.
- This parameter determines the responsiveness of the position loop. A high setpoint shortens the positioning time. Note that an excessively high setpoint may cause vibration.
- The 1st gain set includes C01.00, C01.01, C01.02, and C01.03.

C01.01: 1st speed loop gain

- Defines the speed loop proportional gain.
- This parameter determines the responsiveness of the speed loop. The higher the setpoint, the faster the speed loop response is. Note that an excessively high setpoint may cause vibration.
- In the position control mode, the position loop gain must be increased together with the speed loop gain.

C01.02: 1st speed loop integral time

- Defines the speed loop integral time constant.
- The lower the setpoint, the better the integral action, and the quicker will the deviation value be close to 0.
- There is no integral action when C01.02 is set to 512.00 ms

C01.08: 2nd position loop gain

- Defines the 2nd gain of the position loop.
- The 2nd gain set includes C01.08, C01.09, C01.0A, and C01.0B.
- For details about gain switchover, see section 7.5 "Gain Switchover".

C01.0B: 2nd torque reference filter cutoff frequency

- Defines the torque reference filter time constant.
- Low-pass filtering of torque references helps to smoothen torque references and reduce vibration.
- Pay attention to the responsiveness during setting as an excessively high setpoint lowers down the responsiveness.

NOTICE

- The servo drive offers two low-pass filters for torque references. By default, the 1st filter is used.
- Gain switchover can be used in the position or speed control mode. Once certain conditions are satisfied, the servo drive can switch to filter 2.

C01.11: Cutoff frequency of speed feedback low-pass filter

- Defines the cutoff frequency for first-order low-pass filtering on the speed feedback.
- The lower the setpoint, the weaker the speed feedback fluctuation, and the longer the feedback delay will be.
- Setting this parameter to 8000 Hz negates the filtering effect.

C01.12: Speed feedback overlapping average filter time constant

- Defines the moving average filtering times for speed feedback.
- The higher the setpoint, the weaker the speed feedback fluctuation, but the longer the feedback delay will be.
- When C01.12 is set to a value higher than 0, C01.11 (Cutoff frequency of speed feedback low-pass filter) is invalid.

C01.13: Speed feedforward source

- Defines the source of the speed loop feedforward signal.
- In the position control mode, the speed feedforward control can improve the position reference responsiveness.

Setpoint	Speed Feedforward Source	Remarks
0	No feedforward	-
1	Internal reference	The speed corresponding to the position reference (encoder unit) is defined as the speed feedforward source.
2	Model tracking	Model tracking control can improve the responsiveness and shorten the positioning time. It is only available in the position control mode. It must be used with C02.00. When C02.00 is set to 1, the speed feedforward is sourced from the speed feedforward output of model tracking.
5	Communication	In CSP, 60B1h is used as the source of the external speed feedforward signal. Bit 6 of 607Eh can specify the polarity of the speed feedforward signal (60B1h).

C01.14: Speed feedforward percentage

- In the position control mode, speed feedforward is the value of C01.14 multiplied by the speed feedforward signal, which is part of the speed reference. Increasing the setpoint improves the responsiveness to position references and reduces the position deviation during operation at a constant speed.
- Set C01.15 to a fixed value first, and then gradually increase the value of C01.14 from 0 to a certain setpoint at which speed feedforward achieves the desired effect.
- Adjust C01.15 and C01.14 repeatedly until a balanced setting is achieved.
- For the speed feedforward function and speed feedforward signal selection, see C01.13 (Speed feedforward source selection).

C01.15: Speed feedforward filter cutoff frequency

- Defines the speed feedforward smoothing filter time.

C01.16: Torque feedforward source

- Defines whether to enable the internal torque feedforward function in a non-torque control mode.
- The torque feedforward function can improve the torque reference responsiveness and reduce the position deviation during operation at constant acceleration/deceleration rate.

Setpoint	Torque Feedforward Source	Remarks
0	No feedforward	-
1	Internal reference	The torque feedforward signal source is the speed reference. In the position control mode, the speed reference is output from the position controller. In the speed control mode, the speed reference is output from the user speed reference.
2	Model tracking	It must be used with C02.00. When C02.00 is set to 1, the torque feedforward is sourced from the torque feedforward output of model tracking.
5	Communication	In CSP, 60B1h is used as the source of the external torque feedforward signal. Bit 6 of 607Eh can specify the polarity of the torque feedforward signal (60B1h).

- Torque feedforward parameters include C01.17 (Torque feedforward percentage) and C01.18 (Torque feedforward cutoff frequency).
- In a non-torque control mode, the control block diagram of torque feedforward is as follows:

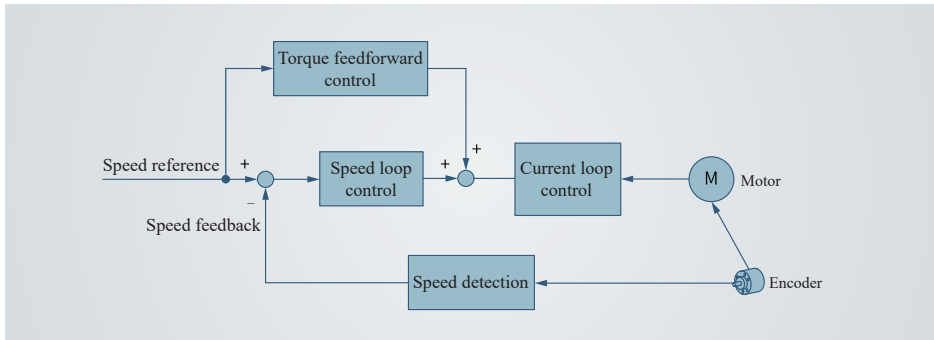


Figure 11-1 Torque feedforward control

C01.17: Torque feedforward percentage

- In control modes other than torque control, torque feedforward is the product of torque feedforward signal multiplied by C01.17 and is part of the torque reference. Increasing the setpoint improves the responsiveness to variable speed references and position references and reduces the position deviation during operation at a constant speed.

C01.18: Torque feedforward filter cutoff frequency

- Defines the filter time constant of torque feedforward.

C01.1B: PDFF control coefficient

- Defines the control method of the speed loop.
- When the setpoint is 100.0, PI control (default control mode of the speed loop) is applied to the speed loop, which features fast dynamic response.
- When the setpoint is 0.0, speed loop integral action is enhanced, which filters out low-frequency interference but also slows down the dynamic response.
- C01.1B can be used to keep a good responsiveness of the speed loop, with the anti-interference capacity in low-frequency bands improved and the speed feedback overshoot not increased.

C01.30: Adaptive notch mode

Setpoint:

- 0: Adaptive notch not updated
- 1: One adaptive notch activated (3rd notch)
- 2: Two adaptive notches activated (3rd and 4th notches)
- 3: Adaptive notch cleared, values of the 3rd and 4th notches restored to default settings
- 4: Resonance point tested only, displayed in C01.31, C01.32, and C01.33

Description:

- Defines the operation mode of the adaptive notch.

C01.38: Gain switchover mode

Setpoint	Gain Switchover Condition	Remarks
0	Fixed to the 1st gain set	The 1st gain set applies.
1	DI switchover	Gains are switched through bit 26 of 60FE. Bit 26 signal inactive: 1st gain set Bit 26 signal active: 2nd gain set If the bit 26 signal cannot be allocated to a DI terminal, the 1st gain set applies.
2	DI P-PI switchover	Gains are switched through bit 26 of 60FE. Bit 26 signal inactive: 1st gain set Bit 26 signal active: 2nd gain set (The 2nd speed loop integral (C01.0A) is forced to be 512 ms.) If the bit 26 signal cannot be allocated to a DI terminal, the 1st gain set applies.
3	Torque reference	When the absolute value of the torque reference exceeds (threshold + loop width, %) in the last 1st gain set, the drive switches to the 2nd gain set. When the absolute value of the torque reference is less than (threshold – loop width, %) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set.
4	Speed reference	When the absolute value of the speed reference exceeds (threshold + loop width, rpm) in the last 1st gain set, the drive switches to the 2nd gain set. When the absolute value of the speed reference is less than (threshold – loop width, rpm) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set.
5	Speed feedback	It is valid only in the position control mode. When the absolute value of the actual speed exceeds (threshold + loop width, rpm) in the last 1st gain set, the drive switches to the 2nd gain set. When the absolute value of the actual speed is less than (threshold – loop width, rpm) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set. The 1st gain set applies when the drive is not in the position control mode.
6	Speed reference change rate	It is valid only in non-speed control modes. When the absolute value of the change rate in the speed reference exceeds (threshold + loop width, 10 rpm/s) in the last 1st gain set, the drive switches to the 2nd gain set. When the absolute value of the change rate in the speed reference is less than (threshold – loop width, 10 rpm/s) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set. The 1st gain set applies in the speed control mode.

Setpoint	Gain Switchover Condition	Remarks
7	Position deviation	When the absolute value of the position deviation exceeds (threshold + loop width, encoder unit) in the last 1st gain set, the drive switches to the 2nd gain set. When the absolute value of the position deviation is less than (threshold + loop width, encoder unit) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set. The 1st gain set applies when the drive is not in the position control mode.
8	Position reference	It is valid only in the position control mode. When the position reference is not 0 in the last 1st gain set, the drive switches to the 2nd gain set. When the position reference is 0 and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set. The 1st gain set applies when the drive is not in the position control mode.

C01.39: Gain switchover time

Defines the duration when the drive switches from the 2nd gain set to the 1st gain set.

C01.3A: Gain switchover threshold

- Defines the gain switchover threshold.
- Gain switchover is affected by both the threshold and the loop width, as defined by C01.38. The unit of gain switchover threshold varies with the switchover condition.
- Set C01.3A to a value greater than or equal to C01.3B. If C01.3A is set to a value less than C01.3B, the servo drive sets C01.3A to the same value as C01.3B.

C01.3B: Gain switchover loop width

- Defines the gain switchover loop width.
- Gain switchover is affected by both the threshold and the loop width. The unit of gain switchover threshold varies with the switchover condition.
- Set C01.3A to a value greater than or equal to C01.3B. If C01.3A is set to a value less than C01.3B, the servo drive sets C01.3A to the same value as C01.3B.

C01.40: Frequency of the 1st notch

- Defines the center frequency of the notch, which is the mechanical resonance frequency.
- In the torque control mode, setting the notch frequency to 8000 Hz deactivates the notch function.

C01.41: Width level of the 1st notch

- Defines the width level of the notch. Use the default value in general cases.
- Width level is the ratio of the notch width to the notch center frequency.

C01.42: Depth level of the 1st notch

- Defines the depth level of the notch.
- The depth level of the notch is the ratio between the input to the output at the notch center frequency.
- The higher the setpoint, the lower the notch depth and the weaker the mechanical resonance suppression will be. Note that an excessively high setpoint may cause system instability.
- For the use of notch, see 7.14 "Vibration Suppression".

C01.45: Depth level of the 2nd notch

- Description of the 2nd notch parameters is the same as that of the 1st notch parameters.

NOTICE

- The 1st and 2nd notches can be set manually or configured as adaptive notches (C01.30 = 1 or 2). In this case, the parameters are automatically set by the drive, while the other three notches can be set manually.

11.3.3 Group C03

C03.21: Speed reference

- It is the speed reference in the local speed mode, which is invalid in EtherCAT mode.

C03.22: Acceleration rate

- It is the acceleration ramp time of the speed reference in the local speed mode, which is invalid in EtherCAT mode.

C03.24: Deceleration rate

- It is the deceleration ramp time of the speed reference in the local speed mode, which is invalid in EtherCAT mode.

C03.27: Internal positive speed limit

- It is the PL of the speed reference in the local speed mode, which is invalid in EtherCAT mode.

C03.28: Internal negative speed limit

- It is the NL of the speed reference in the local speed mode, which is invalid in EtherCAT mode.

C03.43: Internal positive torque limit

- It is valid only in the local torque mode. For torque limit in EtherCAT mode, use 60E0h/60E1h/6072h. Use the torque limit with caution as an excessively low limit value may lead to insufficient motor torque output.
- If the setpoint exceeds the maximum torque of the servo motor and servo drive, the actual torque is limited to the maximum torque of the servo motor and servo drive.

C03.44: Internal negative torque limit

- It is valid only in the local torque mode. For torque limit in EtherCAT mode, use 60E0h/60E1h/6072h.

Use the torque limit with caution as an excessively low limit value may lead to insufficient motor torque output.

- If the setpoint exceeds the maximum torque of the servo motor and servo drive, the actual torque is limited to the maximum torque of the servo motor and servo drive.

C03.47: Positive speed limit in torque mode

- It is valid only in the local torque mode. Use 607F for the speed limit in the EtherCAT, CST, and PT modes.

C03.48: Negative speed limit in torque mode

- It is valid only in the local torque mode. Use 607F for the speed limit in the EtherCAT, CST, and PT modes.

C03.4B: Invalid value for torque reached

- The torque reached function is used to judge whether the actual torque reference reaches the range of the valid value for torque reached. If yes, the servo drive outputs the corresponding flag (bit 10 of the status word) to the host controller.

A: Actual torque reference (U40.02)

B: Base value for torque reach (C03.49)

C: Valid value for torque reach (C03.4A)

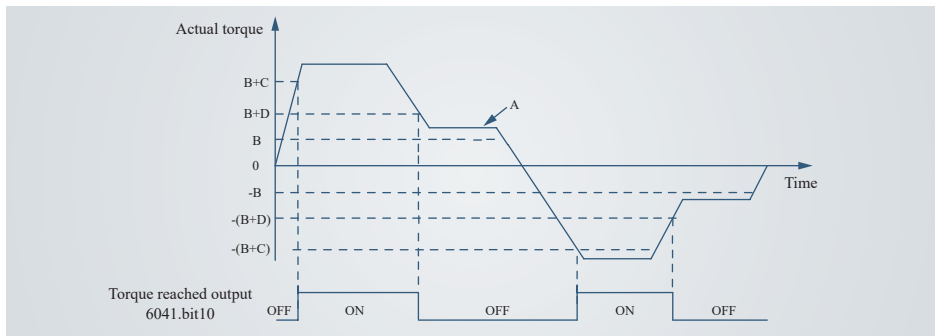
D: Invalid value for torque reach (C03.4B)

C and D are offsets on the basis of B.

The torque reach signal is activated only when the actual torque reference meets the condition: $|A| \geq B + C$

C. Otherwise, the torque reach signal remains inactive.

The torque reach signal is deactivated only when the actual torque reference meets the condition: $|A| < B + D$.



11.3.4 Group C05

C05.0D: Maximum downtime

- Defines the maximum time taken by the motor in decelerating from 6000 RPM to 0 RPM when the stop mode is set to "Ramp to stop as defined by 6084h/609Ah (HM)" or "Ramp to stop as defined by 6085h".

11.3.5 Group C06

C06.04: Input phase loss detection

Servo drives support three-phase 380 V power supplies. When voltage fluctuation or phase loss occurs on the power supply, power input phase loss protection will be triggered by the servo drive based on the setting of C06.04.

- C06.04 = 0: The servo drive reports Er81.0 (Phase loss fault) when the servo drive is set to 3 kW.
- C06.04 = 1: The servo drive does not report Er81.0 (Phase loss fault) when the servo drive is set to 3 kW, with deration of 80%.

C06.11: Motor overload protection threshold

- Determines the motor overload duration before Er41.0 (Motor overload) is reported.
- You can change the setpoint to advance or delay the time when overload protection is triggered based on the motor temperature. The setpoint 50% indicates the time is cut by half; 150% indicates the time is prolonged by 50%.
- Set this parameter based on the actual temperature of the motor.

C06.20: Protection from out of control

- Sets whether to enable the runaway protection function.

11.3.6 Group C0A

C0A.09: Commissioning software communication baud rate

- Defines the communication rate between the servo drive and the host controller.
- The baud rate set in the servo drive must be the same as that in the host controller. Otherwise, communication will fail.

C0A.0A: Commissioning software communication format

- Defines the data check mode between the servo drive and the host controller during communication.
- The data format of the servo drive and the host controller must be the same; otherwise, the communication fails.

11.3.7 Group C13

C13.00: EtherCAT slave name

- Indicates the station number assigned to the slave by the master during EtherCAT communication.

C13.01: EtherCAT slave alias

- Indicates the station number assigned to the slave EtherCAT communication since the master cannot automatically assign station numbers.
- C13.01 = 0: The master assigns the station numbers by default. C13.01 ≠ 0: The set station number applies by default, with the one assigned by master deactivated.

C13.05: EtherCAT synchronization mode setting

- Defines the synchronization work mode:

Setpoint	Function	Remarks
0	Manufacturer Function	Manufacturer Function
1	Sync 1	Applies to the scenarios where the synchronization performance indicator of the host controller jitters for 1 us.
2	Sync 2	Applies to the scenarios where the synchronization performance indicator of the host controller jitters for more than 1 us.

- In the work mode, the synchronization cycle must be an integer multiple of 125 μ s. Otherwise, the servo drive will report Er74.0 (EtherCAT synchronization cycle setting is incorrect.)

C13.06: EtherCAT synchronization error threshold

- Defines the permissible jitter range of synchronization signals when the servo drive works in synchronization mode 1 (C13.05 = 1).

C13.08: EtherCAT enhanced link selection

- When a redundant loop network is used, the EtherCAT Enhanced Link Check function must be enabled (C13.08 = 1), which will take effect upon next power-on of the servo drive.
- When a loop network is used, both C13.08 and C13.19 need to be set to 1.

11.3.8 Group R21R21.00: Drive modelSetpoint:

2: 200EC 3: 400EC 5: 750EC 6: 1000EC

Description:

- Sets the SN of the servo drive. The following table lists the servo drive SNs.

Setpoint	Servo Drive SN	Remarks
2	200EC	The rated drive power is 0.2 kW. The main circuit inputs single-phase 220 V.
3	400EC	The rated drive power is 0.4 kW. The main circuit inputs single-phase 220 V.
5	750EC	The rated drive power is 0.75 kW. The main circuit inputs single-phase 220 V.
6	1000EC	The rated drive power is 1.0 kW. The main circuit inputs single-phase or three-phase 220 V. (The main circuit of the servo drive supports single-phase 220 V power supplies without derating.)

If the voltage input to the main circuit of the servo drive does not comply with the preceding specifications, a fault or damage occurs.

11.3.9 Group F30

F30.03 JOG acceleration/deceleration time

- Acceleration/Deceleration time setpoint for jog in velocity mode, which can be enabled through parameter F30.00 on the panel or through the software

F30.10: Inertia auto-tuning selection

- Used to enable offline inertia auto-tuning through the keypad.
- In the parameter display mode, switch to F30.10 and press the SET key to enable offline inertia auto-tuning. For details about offline inertia auto-tuning, see section 7.2 "Inertia Auto-tuning".

11.3.10 Group F31

F31.00: Fault reset

- Defines whether to enable fault reset.

Setpoint	Function	Remarks
0	No operation	-
1	Enable	When a No.1 or No.2 resettable fault occurs, you can enable the fault reset function in the non-operational state after rectifying the fault cause and stopping the keypad from displaying the fault. When a No.3 warning occurs, you can enable the fault reset function directly.

- For fault classification, see section 10.1.3 "List of faults and alarms".
- The fault reset function, once enabled, stops the keypad from displaying the fault only. It does not activate modifications made on parameters.
- This function is not applicable to non-resettable faults. Use this function with caution in cases where the fault causes are not rectified.

F31.01 Software reset

- Defines whether to enable fault reset.

Setpoint	Function	Remarks
0	No operation	-
1	Enable	Programs in the drive are reset automatically (similar to the program reset upon power-on) after the software reset function is enabled, without the need for a power cycle.

- Software reset conditions: The servo drive is disabled, and there is no non-resettable fault such as No.1 fault.

F31.10: Encoder data reset

- The absolute position saved by the encoder changes abruptly after multi-turn data reset. In this case, perform mechanical homing.

11.3.11 Group U40U40.00: Speed reference

- Indicates the present speed reference (accurate to 1 RPM) of the drive in the position and speed control modes.

U40.01: Speed feedback

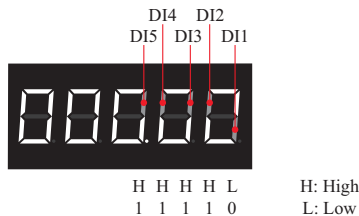
- Indicates the actual motor speed after round-off, which is accurate to 1 rpm.
- This parameter is a 32-bit integer, which is displayed as a decimal on the keypad.

U40.02: Actual torque reference

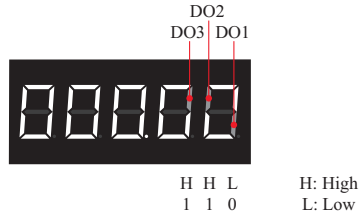
- Indicates the present torque reference (accurate to 0.1%). The value 100.0% corresponds to the rated torque of the motor.

U40.04: DI status

- Indicates the level status of five DIs without filtering.
- Upper LED segments ON: high level (indicated by "1")
Lower LED segments ON: low level (indicated by "0") In cases where DI1 is low level and DI2 to DI5 are high level, the corresponding binary value is 11110, and the value of U40.04 read in the software tool is 30.
- The keypad displays as follows:

U40.05: DO state

- Indicates the level status of three DOs without filtering.
- Upper LED segments ON: high level (indicated by "1")
Lower LED segments ON: low level (indicated by "0") In cases where DO1 is low level and DO2 to DO3 are high level, the corresponding binary value is 110, and the value of U40.05 read in the software tool is 6.
- The keypad displays as follows:



U40.06: Bus voltage

- Indicates the DC bus voltage of the main circuit input voltage after rectification, which is accurate to 0.1 V.

U40.07: Average load ratio

- Indicates the percentage of the average load torque to the rated torque of the motor, which is accurate to 0.1%. The value 100.0% corresponds to the rated torque of the motor.

U40.08: Electrical angle

- Indicates the present electrical angle of the motor, which is accurate to 0.1°.
- The electrical angle variation range is $\pm 360.0^\circ$ when the motor rotates.
- If the motor has four pairs of poles, each revolution generates four rounds of angle changes from 0° to 359.9°.
- Similarly, if the motor has five pairs of poles, each revolution generates five rounds of angle changes from 0° to 359.9°.

U40.09: Mechanical angle

- Indicates present mechanical angle (encoder unit) of the motor. The value 0 indicates that the mechanical angle is 0°.

U40.0C: RMS value of phase current

- Indicates the RMS value of the phase current of the servo motor, which is accurate to 0.1 A.

U40.10: Position deviation counter

- Counts the position pulses fed back by the encoder in any control mode.
- This parameter is a 32-bit integer, which is displayed as a decimal on the keypad.

U40.30: Heatsink temperature

- Indicates the temperature of the module inside the servo drive, which can be used as a reference for estimating the actual temperature of the servo drive.

11.3.12 Group 6000

603Fh: Fault code

- When a fault described in the DSP402 profile occurs on the drive, 603Fh is as described in DSP402.
- When a fault specified by the user occurs on the servo drive, 603Fh is 0xFF00. The value of 603Fh is in hexadecimal.

- In addition, the object dictionary 203Fh displays auxiliary bytes of fault code in hexadecimal.
- 203Fh is a UInt32 value, in which the high 16 bits indicate the internal fault code of the manufacturer, and the low 16 bits indicate the external fault code of the manufacturer.

605Ah: Quick stop option code

- 0: Coast to stop, keeping de-energized status
- 1: Ramp to stop as defined by 6084h/609Ah (HM), keeping de-energized status
- 2: Ramp to stop as defined by 6085h, keeping de-energized status
- 3: Stop at emergency stop torque, keeping de-energized status
- 5: Ramp to stop as defined by 6084h/609Ah (HM), keeping position lock status
- 6: Ramp to stop as defined by 6085h, keeping position lock status
- 7: Stop at emergency stop torque, keeping position lock status

605Ch: Stop mode at S-ON OFF

- 4: Ramp to stop as defined by 6085h, keeping dynamic braking status
- 3: Stop at zero speed, keeping dynamic braking status
- 2: Ramp to stop as defined by 6084h/609Ah (HM), keeping dynamic braking status
- 1: Dynamic braking stop, keeping dynamic braking status
- 0: Coast to stop, keeping de-energized status
- 1: Ramp to stop as defined by 6084h/609Ah (HM), keeping de-energized status

605Dh: Stop option code

- 1: Ramp to stop as defined by 6084h/609Ah (HM), keeping position lock status
- 2: Ramp to stop as defined by 6085h, keeping position lock status
- 3: Stop at emergency stop torque, keeping position lock status

605Eh: Stop mode at No. 2 fault

- 5: Stop at zero speed, keeping dynamic braking status
- 4: Stop at emergency stop torque, keeping dynamic braking status
- 3: Ramp to stop as defined by 6085h, keeping dynamic braking status
- 2: Ramp to stop as defined by 6084h/609Ah (HM), keeping dynamic braking status
- 1: Dynamic braking stop, keeping dynamic braking status
- 0: Coast to stop, keeping de-energized status
- 1: Ramp to stop as defined by 6084h/609Ah (HM), keeping de-energized status
- 2: Ramp to stop as defined by 6085h, keeping de-energized status
- 3: Stop at emergency stop torque, keeping de-energized status
- 4: Dynamic braking stop, keeping de-energized status

6060h: Modes of operation

Setpoint:

- 1: Profile position (PP) mode
- 3: Profile velocity (PV) mode

- 4: Profile torque (PT) mode
- 6: Homing mode (HM)
- 8: Cyclic synchronous position (CSP) mode
- 9: Cyclic synchronous velocity (CSV) mode
- 10: Cyclic synchronous torque (CST) mode
- Others: N/A

Description:

- If an unsupported operation mode is selected through an SDO, an SDO error will be returned.
- If an unsupported operation mode is selected through a PDO, the change of the operation mode will be invalid.

6061h: Modes of operation display

- 1: PP mode
- 3: PV mode
- 4: PT mode
- 6: HM
- 8: CSP mode
- 9: CSV mode
- 10: CST mode

6064h: Position actual value

- Position actual value in user-defined unit (6064h) x Gear ratio (6091h) = Position actual value in encoder unit (6063h)

6065h: Following error window

- When the difference value between position reference (6062h) and position actual value (6064h) keeps exceeding $\pm 6065h$ after the time defined by 6066h elapses, Er47.0 (Position deviation too large) occurs.

6066h: Following error time out

- Defines the time lapse to trigger excessive position deviation, used with 6065h.

6067h: Max. profile velocity

- Defines the threshold for position reach.
- If the difference between the position reference value (6062h) and the position actual value (6064h) is within $\pm 6067h$ and the time reaches 6068h, the position is reached. In this case, bit 10 of 6041h is set to 1 in PP mode.
- This flag bit is meaningful only when the S-ON signal is active in PP mode.

6068h: Position window time

- Defines the window time for position reach, which must be used together with 6067h.

606Dh: Velocity window

- Defines the threshold for speed reach.

- If the difference value between the target speed (60FFh) and the actual speed (606Ch) is within $\pm 606Dh$ and the time reaches 606Eh, the speed is reached and bit 10 of the status word 6041h is set to 1 in the PV mode.
- This flag bit is meaningful only when the S-ON signal is active in PV mode.

606Fh: Velocity threshold

- Defines the threshold for determining whether the user velocity is 0.
- When the velocity actual value (606Ch) is within $\pm 606Fh$ and the time reaches the value set by 6070h, the user velocity is 0. When either condition is not met, the user velocity is not 0.
- This flag bit is valid only in PV mode.
- It is not related to the S-ON state.

6070h: Velocity threshold time

- Defines the time window for determining whether the user velocity is 0, which must be used together with 606Fh.

6071h: Target torque

- Defines the target torque of the servo drive in PT mode.
- The value 1000 corresponds to the rated torque of the motor.

6072h: Max. torque

- Defines the maximum torque reference limit.
- The value 1000 corresponds to the rated torque of the motor.

6074h: Torque reference value

- Defines the target torque value.
- The value 1000 corresponds to the rated torque of the motor.

6077h: Torque actual value

- Indicates the internal torque feedback of the servo drive.
- The value 1000 corresponds to the rated torque of the motor.

607Ah: Target position

- Defines the target position of the servo drive in PP mode.
- When bit 6 of 6040h is set to 0, 607Ah indicates the absolute target position of current segment. After positioning of the current segment is done, the value of 6064h will be the same as the value of 607Ah.
- When bit 6 of 6040h is set to 1, 607Ah indicates the target incremental displacement of the current segment. After positioning of current segment is done, the incremental displacement will be the same as the value of 607Ah.

607Ch: Home offset

- Defines the physical location of mechanical zero that deviates from the home of the motor in position control modes (profile position mode, interpolation mode, and homing mode).

- The home offset in active under the following conditions: The device is powered on, the homing operation is complete, and bit 15 of 6041h is set to 1.
- After homing is done, the position actual value (6064h) will be the same as the value of 607Ch.
- If 607Ch is set to a value outside 607Dh (Software position limit), Er84.3 (Home setting error) will occur.

607D.01h: Minimum software position limit

- Defines the minimum software position limit relative to the mechanical zero.
- Minimum software position limit = (607D.01h)
- The software position limit is used to judge the absolute position. When homing is not performed, the internal software position limit is inactivated.

607D.02h: Maximum software position limit

- Defines the maximum software position limit relative to the mechanical zero.
- Maximum software position limit = (607D.02h)

607Eh: Polarity

- Defines the polarity of position or speed references.
- When bit 7 is 1, it indicates the position reference is multiplied by "-1" and the motor direction is reversed in the standard position mode or interpolation mode.
- When bit 6 is 1, it indicates the speed reference (60FFh) is multiplied by "-1" and the motor direction is reversed in the speed mode.
- When bit 5 is 1, it indicates the torque reference (6071h) is multiplied by "-1" and the motor direction is reversed in the torque mode.
- Other bits are meaningless.

607Fh: Max. profile velocity

- Defines the maximum operating speed in user-defined unit.

6081h: Profile velocity

- Defines the constant operating speed of the target position in PP mode
- The setpoint takes effect after the slave receives the displacement reference.

6083h: Profile acceleration

- Defines the acceleration rate in the acceleration stage of the displacement reference in PP mode.
- The following formula applies if a motor equipped with 17-bit encoder needs to run at 400 RPM (6081h: $400 \times 131072/60$) with acceleration rate being 400 RPM/s (6083h: $400 \times 131072/60$) and deceleration rate being 200 RPM/s (6084h: $200 \times 131072/60$) under a gear ratio of 1:1:
Acceleration time $t_{up} = \Delta 6081h / \Delta 6083h = 1$ (s).
Deceleration time $t_{down} = \Delta 6081h / \Delta 6084h = 2$ (s).
- The setpoint 0 will be forcibly changed to 1.

6084h: Profile deceleration

- Defines the deceleration rate in the deceleration stage of the displacement reference in PP mode.
- The following formula applies if a motor equipped with 17-bit encoder needs to run at 400 RPM (6081h: $400 \times 131072/60$) with acceleration rate being 400 RPM/s (6083h: $400 \times 131072/60$) and deceleration rate being 200 RPM/s (6084h: $200 \times 131072/60$) under a gear ratio of 1:1:

$$\text{Acceleration time } t_{up} = \Delta 6081h / \Delta 6083h = 1 \text{ (s)}$$

$$\text{Deceleration time } t_{down} = \Delta 6081h / \Delta 6084h = 2 \text{ (s)}$$
- The setpoint 0 will be forcibly changed to 1.

6085h: Quick stop deceleration

- Defines the deceleration rate when the quick stop command (6040h set to 0x0002) is active and 605Ah (Quick stop option code) is set to 2 or 5.
- The setpoint 0 will be forcibly changed to 1.

6087h: Torque slope

- Defines the acceleration rate (torque reference increment per second) of the torque reference in PT mode.
- In PT mode, if 605Ah is set to 1, 2, 5, or 6, or 605Dh is set to 1 or 2, the servo drive decelerates to stop as defined by 6087h.
- If the setpoint exceeds the torque reference limit, the limit value will be used.
- The setpoint 0 will be forcibly changed to 1.

6091.01h: Torque slope

- Defines the numerator of the gear ratio.
- The gear ratio is used to establish the proportional relationship between the load shaft displacement designated by the user and the motor shaft displacement.
- The relationship between motor position feedback (encoder unit) and load shaft position feedback (reference unit) is as follows.

$$\text{Motor position feedback} = \text{Load shaft position feedback} \times \text{Gear ratio}$$

The relationship between the motor speed (rpm) and the load shaft speed (reference unit/s) is as follows.

$$\text{Motor speed (rpm)} = \text{Load shaft speed} \times 6091h \times 60 / \text{Encoder resolutions}$$
- The relationship between the motor acceleration (rpm/ms) and the load shaft acceleration (reference unit/s²) is as follows.

$$\text{Motor acceleration (rpm/ms)} = \text{Load shaft acceleration} \times 6091h \times 1000 / \text{Encoder resolutions} / 60$$

6091.02h: Shaft revolutions

- Defines the denominator of the gear ratio.

6098h: Homing method

- For details, see Table 4-1 "Mode lists".

6099.01h: Speed during search for switch

- Defines the speed during search for the deceleration point signal. A large setpoint helps prevent homing timeout.

6099.02h: Speed during search for zero

- Defines the speed in searching for the home signal. Setting this speed to a low value prevents overshoot during stop at high speed, avoiding excessive deviation between the stop position and the set mechanical home.

609Ah: Homing acceleration

- Defines the acceleration rate in homing mode.

60B8h: Touch Probe Function

Bit	Name	Description
0	Touch probe 1 function selection 0: Disabled 1: Enabled	bit0 to bit5: Probe 1 related settings When DI is used as the probe trigger signal, the DI source cannot be changed after the probe is enabled. For an absolute encoder, the Z signal refers to the zero point of motor single-turn position feedback.
1	Touch probe 1 trigger mode 0: Single trigger mode (Latches the position at the first trigger event.) 1: Continuous trigger	
2	Touch probe 1 trigger signal selection 0: DI signal 1: Z signal	
3	Reserved	
4	Touch probe 1 positive edge 0: Switch off latching at positive edge 1: Enable latching at positive edge	
5	Touch probe 1 negative edge 0: Switch off latching at negative edge 1: Enable latching at negative edge	
6 to 7	Reserved	
8	Touch probe 2 function selection 0: Disabled 1: Enabled	bit8 to bit13: Probe 2 related settings
9	Touch probe 2 trigger mode 0: Single trigger mode (Latches the position at the first trigger event.) 1: Continuous trigger	
10	Touch probe 2 trigger signal selection 0: DI signal 1: Z signal	
11	Reserved	

Bit	Name	Description
12	Touch probe 2 positive edge 0: Switch off latching at positive edge 1: Enable latching at positive edge	bit8 to bit13: Probe 2 related settings
13	Touch probe 2 negative edge 0: Switch off latching at negative edge 1: Enable latching at negative edge	
14 to 15	Reserved	

60BAh: Touch probe 1 positive edge

- Indicates the position feedback value (reference unit) latched at positive edge of touch probe 1 signal.

60BBh: Touch probe 1 negative edge

- Indicates the position feedback value (reference unit) latched at negative edge of touch probe 1 signal.

60BCh: Touch probe 2 positive edge

- Indicates the position feedback value (reference unit) latched at positive edge of touch probe 2 signal.

60BDh: Touch probe 2 negative edge

- Indicates the position feedback value (reference unit) latched at negative edge of touch probe 2 signal.

60C5h: Max. acceleration

- Defines the maximum permissible deceleration in PP mode, PV mode, and homing mode.
- The setpoint 0 will be forcibly changed to 1.

60C6h: Max. deceleration

- Defines the maximum permissible deceleration in PP mode, PV mode, and homing mode.
- The setpoint 0 will be forcibly changed to 1.

60D5h: Touch probe 1 positive edge counter

- The counting value is added by "1" each time this object is triggered.

60D6h: Touch probe 1 negative edge counter

- The counting value is added by "1" each time this object is triggered.

60D7h: Touch probe 2 positive edge counter

- The counting value is added by "2" each time this object is triggered.

60D8h: Touch probe 2 negative edge counter

- The counting value is added by "2" each time this object is triggered.

60E0h: Positive torque limit

- Defines the maximum torque limit of the servo drive in the forward direction.

60E1h: Negative torque limit

- Defines the maximum torque limit of the servo drive in the reverse direction.

60E3.01h: 1st supported homing method

- Bit 0 to bit 7: The low 8 bits indicate the supported homing method. Set 6098h to the corresponding value.
- Bit 8: Relative position homing
0: Not supported
1: Supported
- Bit 9: Absolute position homing
0: Not supported
1: Supported
- Bit 10 to bit 15: N/A

60E6h: Actual position calculation method

- Defines the method for calculating the mechanical position after homing is completed. After homing is triggered, changes in 60E6h will be blocked.

60F4h: Position deviation

- This object indicates the position deviation (in reference unit).

60FCh: Position reference value

- Indicates the position reference (encoder unit).
- If no warning is detected when the S-ON signal is active, the relationship between the position reference in reference unit and that in encoder unit is as follows:

$$60FCh \text{ (in encoder unit)} = 6062h \text{ (in reference unit)} \times 6091h$$

60FDh: DI status

- Indicates current DI logic of the drive. 0: Inactive; 1: Active

Bit	Description
0	Reverse overtravel active
1	Forward overtravel active
2	Home signal active
3 to 15	N/A
16	DI1 input active
17	DI2 input active
18	DI3 input active
19	DI4 input active

Bit	Description
20	DI5 input active
21 to 26	NA

60FFh: Target velocity

- Defines the target velocity in CSV and PV mode.

6502h: Supported drive modes

- Defines the target velocity in CSV and PV mode.